

24 Electrochemistry

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

The Nernst equation accounts for concentrations in electrode potential; E changes with reaction quotient rather than just the standard value. For a spontaneous cell reaction $\Delta G = -nFE$, with n the electrons transferred. Electrolysis requires charge $Q = It$ and amount electrons Q/F ; electron stoichiometry determines product amount.

Worked example: follow the reasoning

At constant current 2.0 A for 965 s, $Q = 1930 \text{ C} \approx 0.020 \text{ mol electrons}$ ($F \approx 96500 \text{ C mol}^{-1}$).

Quick check · multiple choice

A How many moles electrons deposit 1 mol Cu from Cu^{2+} ? [1]

- A 0.5
- B 1
- C 3
- D 2

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Copper is deposited by $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ using 1.0 A for 1930 s. Find amount of Cu formed. [3]

Final answer: _____

Answers and explanation

Quick check: D. 2

Question 1 [3 marks]

$Q = It = 1930 \text{ C}$ [1]; electrons = $1930/96500 = 0.020 \text{ mol}$ [1]; Cu = 0.010 mol [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.